Note 3.4 Basis and Dimension

Basis

Definition of Basis

The vectors $\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n$ form a basis for a vector space V if and only if

- $\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly independent.
- $\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n$ span V.
  That is, the basis is the spanning set without redundant vectors.

Theorem 3.4.1

If $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a spanning set for a vector space V, then any collection of m vectors in V, where $m > n$, is linear dependent.

Proof:

Let $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ be the vectors in V, where $m > n$. Then they can be represented as

$$\mathbf{u}_i = a_{i1}\mathbf{v}_1 + a_{i2}\mathbf{v_2} + \cdots + a_{in}\mathbf{v}_n, 1 \leq i \leq m$$

Then the linear combination of $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ equals to zero is

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m = \mathbf{0}$$

Substitution, we get

$$\sum_{k=1}^{m}\left(c_k \cdot \sum_{j=1}^{n} a_{kj}\mathbf{v}_j\right) = \sum_{j=1}^{n}\left(\sum_{k=1}^{m} a_{kj}c_k\right)\mathbf{v}_j = \mathbf{0}$$

Now we can let the coefficients all equal to zero to get the system of equations. (Here, it is not important for whether the spanning set is basis)

$$\sum_{k=1}^{m} a_{kj}c_k = 0, j = 1, 2, \cdots, n$$

This is a homogeneous system with more unknowns than equations. By theorem 1.2.1, there must exist a nontrivial solution $(\hat{c_1}, \hat{c_2}, \cdots, \hat{c_n})$. Then the linear combination of $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ equals to zero exists nontrivial solution. So they are linearly dependent.

Corollary 3.4.2

If both $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\}$ are bases for a vector space V, then $n = m$.

Dimension

Definition of Dimension

Let V be a vector space. If V has a basis consisting of n vectors, we say that V has dimension n. And the subspace $\{\mathbf{0}\}$ of V is said to have dimension 0. V is said to be finite dimensional if there is a finite set of vectors that spans V; otherwise we say it is infinite dimensional.

Theorem 3.4.3

If V is a vector space of dimension $n > 0$, then any n vectors can span V if and only if they are linearly independent.

Proof :

From left to right:
Suppose $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is the set of any n vectors that span V. If they are linearly dependent, there exists at least one vector in the set can be represented as the linear combination of the other vectors. Then it will be less than n vectors span the V, which is conflict with $\dim V = n$.
From right to left:
Suppose $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is the set of any n vectors and $\mathbf{v}$ is any other vector in V. Since $\dim V = n$, then the basis consists of n vectors which span V. Then by the Theorem 3.4.1, the vectors $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n, \mathbf{v}\}$ must be linear dependent. Then there exist

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n + c_{n+1}\mathbf{v} = \mathbf{0}$$

Since $\{\mathbf{v_1}, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is linear independent, $c_{n+1} \neq 0$. Then

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$$

where $a_i = \dfrac{c_i}{c_{n+1}}$. So every vectors $\mathbf{v}$ can be expressed as the linear combination of the n vectors. So they span V.

Theorem 3.4.4

If V is a vector space of dimension $n > 0$, then

- no set of fewer than n vectors can span V.
- any subset of fewer than n linearly independent vectors can be extended to form a basis for V.
- any spanning set containing more than n vectors can be pared down to form a basis for V.

Standard Bases

We refer to the set $\{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_n\}$ as the standard basis for $\mathbb{R}^3$.